



Cairo University
 Faculty of Engineering
 Credit Hours System
 Communications and Computer Engineering Department

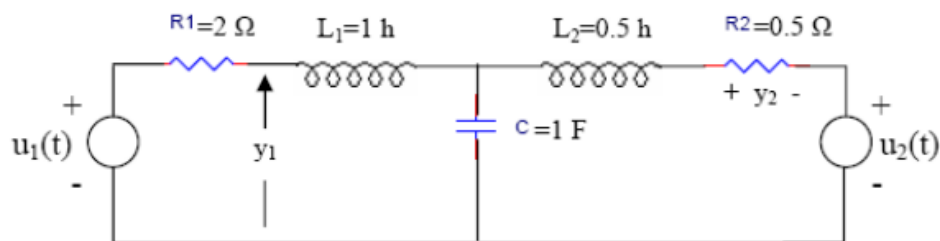
Fall 2026

Course Code EECS 304
 Course Title Control I
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Sheet 8: State Space Representation

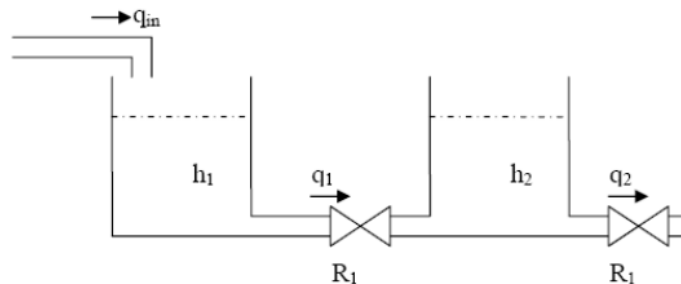
Question (1):

For the following networks write down the state equations and output equations and get the corresponding state space representation.



Question (2):

For the interacting two tank system shown below derive a suitable state space representation if the system has two outputs, the first output is the liquid level in the second tank and the second output is the output flow rate from the second tank.



Question (3):

For the system:

$$\dot{\underline{x}} = \begin{bmatrix} -5 & -1 \\ 3 & -1 \end{bmatrix} \underline{x} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u, \quad y = \begin{bmatrix} 1 & 2 \end{bmatrix} \underline{x}$$

Obtain the T.F. of the system

Question (4):

Find a phase state space representation for the following systems:

1. $\frac{Y(s)}{U(s)} = \frac{1}{(s+1)(s+3)}$

2. $\frac{Y(s)}{U(s)} = \frac{1}{2s^4 + s^3 + s^2 + 27s + 50}$

3. $\frac{Y(s)}{U(s)} = \frac{10s + 1}{2s^3 + s^2 + s + 10}$

4. $\ddot{y} + 5\dot{y} + 3y = 2\dot{f}(t) + f(t)$

Question (5):

A certain system is described by the following set of differential equations:

$$\ddot{y}_1 + \dot{y}_1 + 2y_1 - 2y_2 = u_1$$

$$\ddot{y}_2 - y_1 + y_2 = u_2$$

1. Write the state and output equations of the system.
2. Find the transfer function matrix between the outputs and inputs.